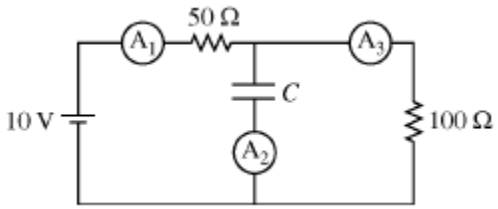
Chapter 18 Bookwork
CH 18 WS #3

Name: _____
Per: _____

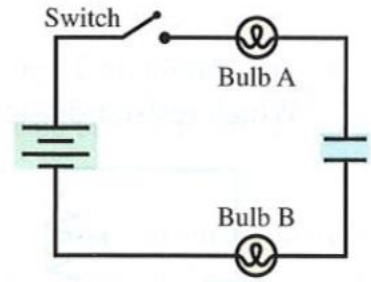
Conceptual –

1. A 20 kΩ resistor is connected in series with an initially uncharged 100 μF capacitor and a 5 V battery. What is the charge on the capacitor when the circuit has reached steady state? (**Pick One**): A) 5×10^{4} C; B) 2.5×10^{-1} C; C) 2.5×10^{-4} C; D) 5×10^{-4} C

The figure shows a 10 V battery connected in a circuit with two resistors, a parallel-plate capacitor of capacitance C, and three ammeters. The circuit has been connected for a long time.



2. Let I_i be the current in ammeter A_i. Which of the following correctly ranks the currents in the three ammeters? (**Pick One**): A) $(I_1 = I_3) > I_2$; B) $I_1 > (I_3 = I_2)$; C) $I_1 > I_3 > I_2$; D) $I_3 > (I_1 = I_2)$
3. Which of the following is correct about the voltage Vc across the capacitor? (**Pick One**): A) $V_C = 0$; B) $0 < V_C < 10$ V; C) $V_C = 10$ V; D) $V_C > 10$ V
4. Assume the capacitance C of the capacitor is known. The gap between the capacitor's plates is now filled with an unknown insulating material, and the circuit is again left connected for a long time. Measuring which of the following provides enough information to determine the unknown material's dielectric constant κ? (**Pick One**): A) The potential difference across the capacitor and the current in each branch of the circuit; B) The potential difference across the capacitor and the charge on one of its plates; C) The potential difference across the capacitor and the charge on the dielectric material; D) The separation between the capacitor plates and the area of the plates.
5. The figure shows a circuit consisting of a battery, a switch, two identical lightbulbs, and a capacitor that is initially uncharged.

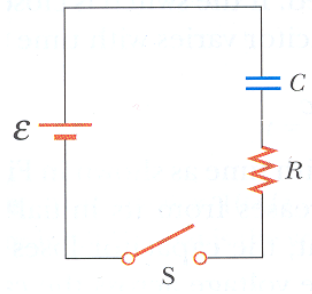
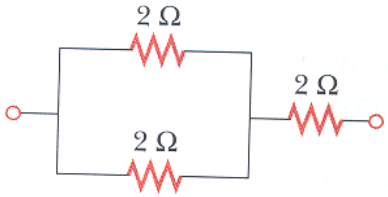


   a. Immediately after the switch is closed, are either or both bulbs glowing? Explain.
   b. If both bulbs are glowing, which is brighter? Or are they equally bright? Explain.
   c. For any bulb (A or B or both) that lights up immediately after the switch is closed, does its brightness increase with time, decrease with time, or remain unchanged? Explain.
6. Two students measure the time constant of an RC circuit. The first student charges the capacitor using a 12 V battery, then lets the capacitor discharge the resistor. The second student repeats the experiment using a 5 V battery. Which student measures the longer time constant? Explain.

Problems –

7. Consider a series *RC* circuit for which C = 6.0 μF, R = 2.0×10^{6} Ω, and ε = 20 V. Find a) the time constant of the circuit and (b) the maximum charge on the capacitor after a switch in the circuit is closed. (12 s, 120 μC)

8. An uncharged capacitor and a resistor are connected in series to a source of emf. If $\varepsilon = 9.0$ V, C = 20 μF, and R = 100 Ω, find (a) the time constant of the circuit, (b) the maximum charge on the capacitor, and (c) the charge on the capacitor after one time constant.
(2.0 x 10^{-3} s, 180 μC, 114 μC)
9. Consider a series RC circuit (see figure) for which R = 1.0 MΩ, C = 5.0 μF, and $\varepsilon = 30$ V. Find the charge on the capacitor 10 s after the switch is closed. (130 μC)



10. An uncharged capacitor, C = 10μF, and a resistor are connected in series to an emf of 9.0 V. (a) What should the resistance be to give a time constant of 5.0s? (b) What is the charge on the capacitor after two time constants have elapsed, if the resistance has the value found in part (a)? (5.0 x 10^5 Ω, 78 μC)
11. Show that $\tau = RC$ has units of time.
12. An electric heater is rated at 1300 W, a toaster is rated at 1000 W, and an electric grill is rated at 1500 W. The three appliances are connected in parallel to a common 120-V circuit. (a) How much current does each appliance draw? (b) Is a 30.0-A circuit breaker sufficient in this situation? Explain. (10.8 A, 8.33 A, 12.5 A, no)
13. Your toaster oven and coffee maker each dissipate 1200 W of power. Can you operate them together if the 120-V line that feeds them has a circuit breaker rated at 15 A? Explain. (no…make sure to explain)
14. A heating element in a stove is designed to dissipate 3000 W when connected to 240 V. (a) Assuming that the resistance is constant, calculate the current in this element if it is connected to 120 V. (b) Calculate the power it dissipates at this voltage.
(6.25 A, 750 W)
15. Three 2.0-Ω resistors are connected as shown in the figure. Each can dissipate a maximum power of 32 W without being excessively heated. Determine the maximum power the network can dissipate. (48 W)